

Can the AI Reading Step Be Reproduced?

Seven seed papers, read three times each by a scripted `claude -p` reader

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Summary

Every literature link in the POTS ontology starts as a claim that an AI reader copied from a seed paper. In the first run, those readers were interactive Claude Code helpers. We recorded what they were given, but we could not replay the calls. So we turned the reader into a script that gives every call exactly the same inputs. We ran it three times on each of the seven papers and measured how far the readings agree with each other and with the current ontology. The calls ran on the lab's Claude subscription, without an API key, and were not billed separately.

Overlap between repeats (whole ontology)	86%	Calls	21 (7 papers $\times$ 3)
Edges found by all 3 repeats	316 of 395 (80%)	Failed calls	0
Current ontology links reproduced by the majority	115 of 132 (87%)	Time	62 min of calls (4 at once)
Quotes verbatim in the paper	1,345 of 1,368 (98%)	Tokens	0.92M in, 0.44M out
Verdict identical across all 3 repeats	270 of 316 (85%)	Plan use	+2% of the weekly limit

Table 1: Headline numbers. An *edge* is one (subject, predicate, object) triple claimed by one paper.

Answer. The readings are not byte-identical, because Claude Code does not expose temperature or a seed. But they are stable: about 4 in 5 edges appear in every repeat, and a majority vote over three repeats recovers 87% of the links we have now. Reproducibility therefore comes from fixed, hashed inputs, repeated readings and a majority rule, not from identical text.

1 How each call is pinned down

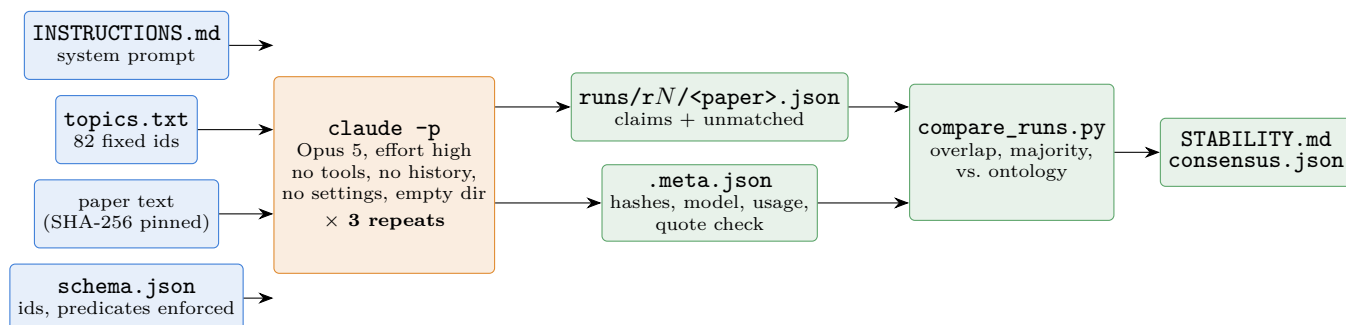


Figure 1: The scripted reader. Blue: fixed inputs. Orange: the only step that is not deterministic. Green: stored and replayed offline (`compare_runs.py -check` runs in the tests).

Fixed and recorded	Instructions, topic list, paper text, output schema, model id, effort, Claude Code version (2.1.287). A call is skipped on re-run only if all its input hashes match, so an interrupted run resumes and a finished one costs nothing.
Closed off	Tools, session history, user and project settings and hooks, the project's <code>CLAUDE.md</code> , and <code>ANTHROPIC_API_KEY</code> (dropped so a call cannot bill an API account).
Enforced by the schema	Subject and object must be one of the 82 vocabulary ids, the predicate one of nine, and the verdict <i>supports</i> or <i>partial</i> . Readers cannot invent topics: anything missing goes into <code>unmatched</code> .
Not controllable	Sampling temperature and seed. This is why we repeat each reading and vote.

Table 2: What the script controls.

Why Opus 5, not Opus 5.5. On a trial call, Opus 5.5's biology safeguards stopped the reading of the Larsen long-COVID abstract. This was a false positive, triggered by virology vocabulary. Claude Code then handed the session to Opus 5 without failing the call; in one case it returned an empty answer instead. A run that silently mixes two models cannot be reproduced. So the script pins Opus 5, and it rejects any call whose usage names another model or that returns no claims. None of the 21 calls was rejected.

2 How much the repeats agree

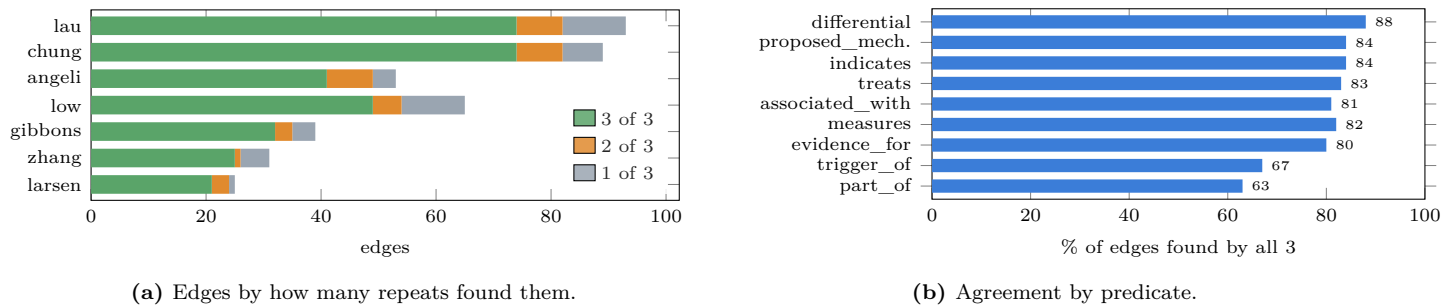


Figure 2: Run-to-run agreement. Lau and Low, the two longest narrative reviews, have the most one-off edges. Battery membership (`part_of`) and triggers are the least stable.

Paper	Claims per repeat	Edges per repeat	Overlap	Majority	In ontology	Recall	Majority, not in ontology
Lau 2026 (review)	101, 100, 111	84, 80, 85	86%	82	64	89%	25
Chung & Raj 2026 (review)	89, 94, 91	82, 81, 82	89%	82	81	88%	11
Angeli 2024	51, 57, 54	49, 45, 49	85%	49	42	90%	11
Low 2009	58, 60, 73	53, 55, 60	83%	54	57	86%	5
Gibbons 2013	36, 42, 40	34, 36, 36	88%	35	25	96%	11
Zhang 2022	32, 33, 34	26, 27, 29	87%	26	21	95%	6
Larsen 2025 (abstract)	25, 23, 22	25, 23, 22	89%	24	16	81%	11

Table 3: By paper. *Overlap*: mean pairwise Jaccard. *Majority*: found by at least 2 of 3. *Recall*: the share of that paper’s current links that the majority also has.

3 Where the scripted reader and the ontology differ

The current ontology rests on a single reading of each paper by the first run’s readers, followed by curation. So a difference is a disagreement, not proof that the new reader is wrong.

Kind	<i>n</i>	Pattern
In the ontology, missed by the majority	17	Mostly indirect, single-paper inferences: <code>postinfectious</code> → <code>autoimmune/neuropathic</code> , <code>systemic_autoimmune_disease</code> → <code>autoimmune</code> or <code>neuropathic</code> , treatments <i>indicating</i> a cause (<code>volume_expansion_therapy</code> → <code>neuropathic</code>), and <code>ambulatory_monitoring</code> links. Two of these were found by one repeat only.
In the majority, not in the ontology	28	6 are <code>part_of</code> <code>orthostatic_challenge_testing</code> . The first run’s instructions did not offer that battery as an object, so its readers could not use it. The rest are plausible claims the first run missed or merged: <code>long_covid</code> <code>trigger_of</code> <code>postinfectious_evidence</code> , <code>prokinetic_therapy</code> <code>treats</code> <code>gi_dysmotility</code> , <code>functional_neurological_disorder</code> as a <code>differential</code> .

Table 4: Disagreements with the current ontology (full lists in `extraction/2026-10-02/STABILITY.md`).

Quotes. 23 of 1,368 quotes (1.7%) failed the verbatim check, and they were the same 7 passages each time. In 6 of them the reader quoted the sentence as printed and skipped a figure caption or running page header that the PDF text layer had inserted mid-sentence. That is a limitation of the checker, not an invented quote. In 1 (Low 2009) the reader joined two sentences that are not adjacent, which breaks the rules.

Missing topics. Because readers could not invent ids, they listed 114 topics the vocabulary lacks. The ones that recur across repeats include microvascular dysfunction, Chiari malformation, mitral valve prolapse, Raynaud’s phenomenon, bupropion, and phosphorylated α -synuclein on skin biopsy. Each is a candidate for the next vocabulary review.

4 What this means and what comes next

- **The method holds up.** Three independent scripted readings agree on 86% of links and recover 87% of the current ontology. Reproducibility is something we measure, and anyone with the texts can re-run it in about an hour of calls.
- **Adopt the majority rule.** A link enters the ontology only if at least 2 of 3 readings of the same paper find it. Single-repeat edges (11% of edges) become a review queue, not links.
- **Remaining gaps.** No clinician gold standard yet, so agreement measures stability, not correctness. Next step: have a clinician label the 17 + 28 disagreements. In addition, the checker should tolerate page furniture in PDF text, and the Low 2009 stitched quote should be rejected automatically.
- **Cost.** The calls ran on the subscription: the weekly limit went from 32% to 34%. The \$17 shown in the logs is a list-price equivalent, not a charge.

Files. `scripts/extract_claims.py`, `scripts/compare_runs.py`, `extraction/2026-10-02/` (inputs, 21 outputs, `STABILITY.md`). The ontology itself is unchanged; it is still assembled from the 2026-10-01 run.